

Day 25: RIP & EIGRP

ROUTING INFORMATION PROTOCOL (RIP)

- Routing Information Protocol (Industry Standard)
- is a DISTANCE VECTOR IGP
 - uses Routing-By-Rumor logic to learn/share routes
- Uses HOP COUNT as it's METRIC (One Router = One Hop) Bandwidth is irrelevant
- MAX HOP COUNT is 15 (anything more is considered unreachable)
- Has THREE VERSIONS:
 - RIPv1 and RIPv2; used for IPv4
 - RIPv6 (RIP Next Generation) used for IPv6
- Uses TWO MESSAGE TYPES:
 - REQUEST :
 - To ask RIP-ENABLED neighbour ROUTERS to send their ROUTING TABLE
 - RESPONSE:
 - To SEND the LOCAL router's ROUTING TABLE to neighbouring ROUTERS

By DEFAULT, RIP-Enabled ROUTERS will share their ROUTING TABLE every 30 seconds

RIPv1 and RIPv2

RIPv1:

- Only advertises *classful addresses* (Class A, Class B, Class C)
- Doesn't support VLSM, CIDR
- Doesn't include SUBNET MASK information in ADVERTISEMENTS (RESPONSE messages)

- Example:

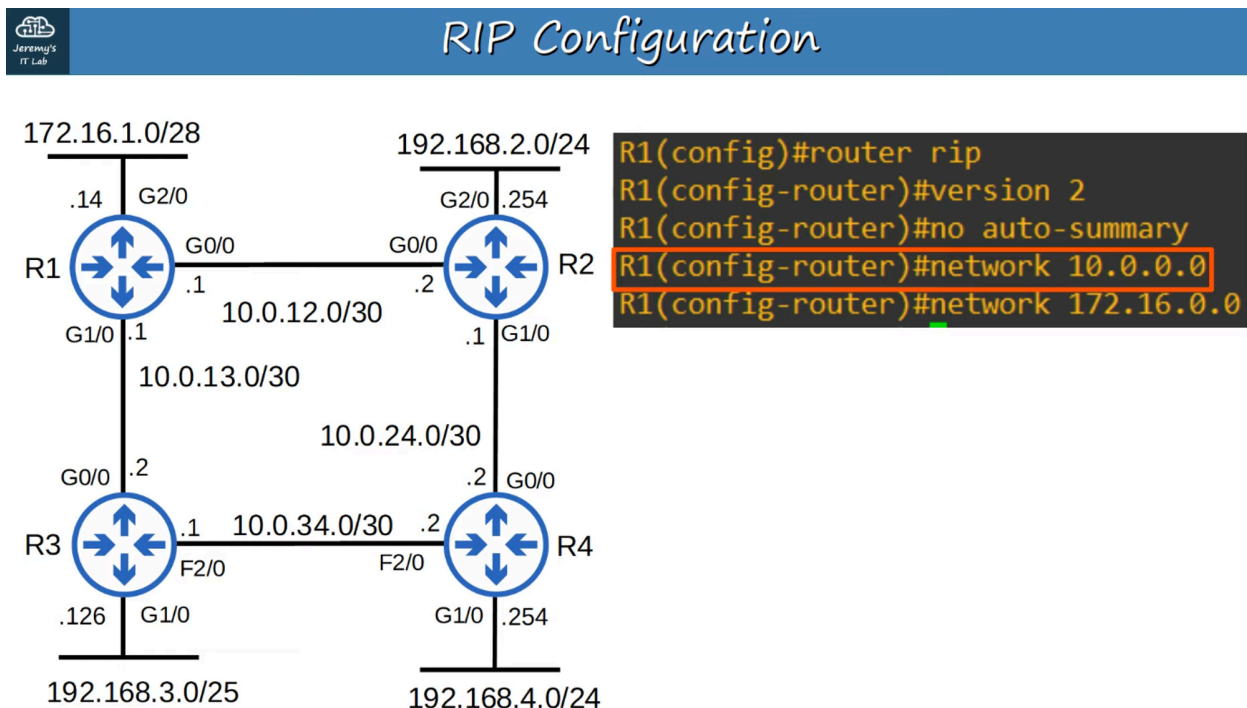
- 10.1.1.0/24 will become 10.0.0.0 (Class A Address, so assumed to be /8)
- 172.16.192.0/18 will become 172.16.0.0 (Class B Address, so assumed to be /16)
- 192.168.1.40/30 will become 192.168.1.0 (Class C Address, so assumed to be /24)

- Messages are BROADCAST to 255.255.255.255

RIPv2:

- Supports VLSM, CIDR
- Includes SUBNET MASK information in ADVERTISEMENTS
- Messages are **multicast** to 224.0.0.9
 - Broadcast Messages are delivered to ALL devices on the local network
 - Multicast Messages are delivered only to devices to have joined that specific **multicast group**

CONFIGURING RIP



The **"network"** command tells the router to:

- Look for INTERFACES with an IP ADDRESS that is in the specific RANGE
- ACTIVATES RIP on the INTERFACES that fall in the RANGE
- Form ADJACENCIES with connected RIP neighbors
- Advertise the **NETWORK PREFIX of the INTERFACE** (NOT the prefix in the "network" command)

The OSPF and EIGRP **"network"** commands operate in the same way

Because the RIP "network" command is CLASSFUL. It will automatically convert to CLASSFUL networks

- 10.0.0.0 is assumed to be 10.0.0.0/8
- R1 will look for ANY INTERFACES with an IP ADDRESS that matches 10.0.0.0/8 (because it is /8 it only needs to match the FIRST 8 bits)
- 10.0.12.1 and 10.0.13.1 both match SO RIP is ACTIVATED on G0/0 and G0/1
- R1 then forms ADJACENCIES with its neighbors R2 and R3
- R1 ADVERTISES 10.0.12.0/30 and 10.0.13.0/30 (NOT 10.0.0.0/8) to it's RIP neighbors

```
R1(config-router)#network 172.16.0.0
```

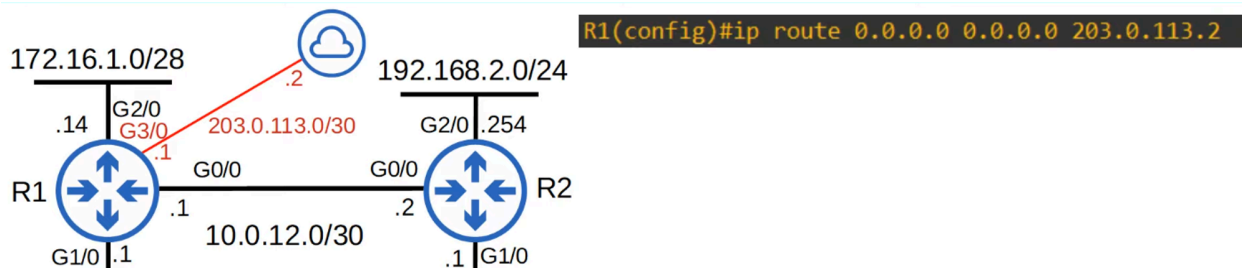
- Because the "network" command is CLASSFUL, 172.16.0.0 is assumed to be 172.16.0.0/16
- R1 will look for ANY INTERFACES that match 172.16.0.0/16
- 172.16.1.14 matches, so R1 will ACTIVATE RIP on G2/0
- There are NO RIP neighbors connected to G2/0 so no NEW ADJACENCIES are formed
 - Although there are NO RIP neighbors, R1 will still send ADVERTISEMENTS out of G2/0.

- This is unnecessary traffic, so G2/0 should be configured as a **passive interface**

```
R1(config-router)#passive-interface g2/0
```

- the “passive-interface” command tells the ROUTER to stop sending RIP advertisements out of the specified interface (G2/0)
- However, the ROUTER will continue to ADVERTISE the network prefix of the interface (172.16.1.0/28) to it's RIP neighbors (R2, R3)
- You should ALWAYS use this command on INTERFACES which don't have any RIP neighbors
- EIGRP and OSPF both have the same passive INTERFACE functionality, using the same command.

HOW TO ADVERTISE A DEFAULT ROUTE INTO RIP



```
Gateway of last resort is 203.0.113.2 to network 0.0.0.0
S* 0.0.0.0/0 [1/0] via 203.0.113.2
```

To SHARE this DEFAULT ROUTE with R1's RIP neighbors, using this command:

```
R1(config-router)#default-information originate
```

RIP doesn't care about interface AD cost (RIP cost is 120), only “hops”.

Since both have an equal number of "hops", both paths appear in the DEFAULT ROUTE (Gateway of Last Resort)

```
Gateway of last resort is 10.0.34.1 to network 0.0.0.0

R*    0.0.0.0/0 [120/2] via 10.0.34.1, 00:00:06, FastEthernet2/0
        [120/2] via 10.0.24.1, 00:00:01, GigabitEthernet0/0
```

"show ip protocols" (for RIP)

```
R1#show ip protocols
*** IP Routing is NSF aware ***

Routing Protocol is "rip"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Sending updates every 30 seconds, next due in 28 seconds
  Invalid after 180 seconds, hold down 180, flushed after 240
  Redistributing: rip
  Default version control: send version 2, receive version 2
    Interface          Send  Recv  Triggered RIP  Key-chain
  GigabitEthernet0/0    2     2
  GigabitEthernet1/0    2     2
  Automatic network summarization is not in effect
  Maximum path: 4
  Routing for Networks:
    10.0.0.0
    172.16.0.0
  Passive Interface(s):
    GigabitEthernet2/0
  Routing Information Sources:
    Gateway          Distance      Last Update
    10.0.12.2         120          00:00:21
    10.0.13.2         120          00:00:06
  Distance: (default is 120)
```

"Maximum path: 4" is the DEFAULT but can be changed with this command:

```
R1(config-router)#maximum-paths ?  
  <1-32>  Number of paths
```

```
R1(config-router)#maximum-paths 8
```

"Distance" (AD) can be changed with this command (DEFAULT is 120)

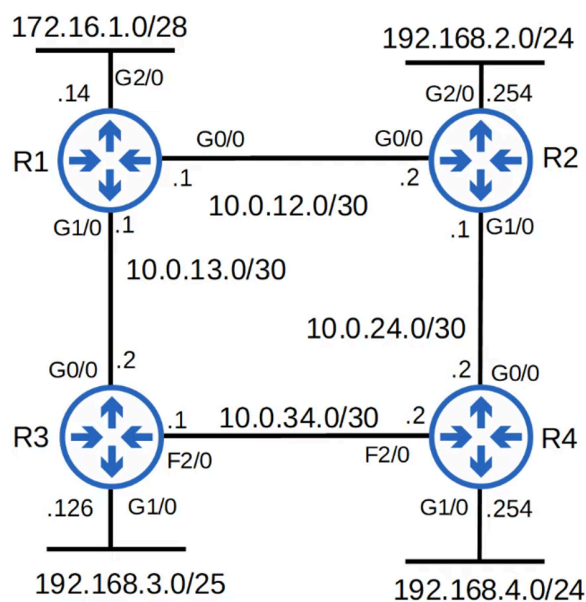
```
R1(config-router)#distance ?  
  <1-255>  Administrative distance
```

```
R1(config-router)#distance 85
```

ENHANCED INTERIOR GATEWAY ROUTING PROTOCOL (EIGRP)

- Enhanced Interior Gateway Routing Protocol
 - is a DISTANCE VECTOR IGP
 - Was Cisco proprietary, but Cisco has now published it openly so other vendor can implement it on their equipment
 - Considered an "advanced" / "hybrid" DISTANCE VECTOR ROUTING PROTOCOL
 - Much faster than RIP in reacting to changes in the NETWORK
 - Does NOT have the 15 'hop count' limit of RIP
 - Sends messages using MULTICAST ADDRESS **224.0.0.10 (Memorize this number)**
 - Is the ONLY IGP that can perform **unequal** cost load-balancing (by DEFAULT, it performs ECMP load-balancing over 4 paths like RIP)
-

CONFIGURATION OF EIGRP



```
R1(config)#router eigrp 1
R1(config-router)#no auto-summary
R1(config-router)#passive-interface g2/0
R1(config-router)#network 10.0.0.0
R1(config-router)#network 172.16.1.0 0.0.0.15
```

"router eigrp "

- The AS (Autonomous System) number MUST MATCH between ROUTERS or they will NOT form an ADJACENCY and share ROUTE information
- Auto-summary might be ENABLED or DISABLED by DEFAULT; depending on the ROUTER/IOS version. If ENABLED, DISABLE it.
- The **"network"** command will assume a CLASSFUL ADDRESS, if you don't specify the SUBNET MASK
- EIGRP uses a *wildcard mask* instead of a regular subnet mask

A WILDCARD MASK is an "inverted" SUBNET MASK

- All 1's in the SUBNET MASK are 0 in the equivalent WILDCARD MASK.
- All 0s in the SUBNET MASK are 1 in the equivalent WILDCARD MASK.

Wildcard masks

- A shortcut is to subtract each octet of the subnet mask from 255.

```

1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 1 1 1 0 0 0 . 0 0 0 0 0 0 0 0
255      .      255      .      248      .      0
255 - 255 ↓      255 - 255 ↓      255 - 248 ↓      255 - 0 ↓
0 0 0 0 0 0 0 0 . 0 0 0 0 0 0 0 0 . 0 0 0 0 0 1 1 1 . 1 1 1 1 1 1 1 1
0      .      0      .      7      .      255

```

/21

"0" in the WILDCARD MASK = BITS MUST MATCH !

"1" in the WILDCARD MASK = Do not have to match

Wildcard masks

- "0"
- "1"

Match! EIGRP will be activated on the interface.

```

1 0 1 0 1 1 0 0 . 0 0 0 1 0 0 0 0 . 0 0 0 0 0 0 0 1 . 0 0 0 0 1 1 1 0
172      .      16      .      1      .      14

```

EIGRP network command:

```

1 0 1 0 1 1 0 0 . 0 0 0 1 0 0 0 0 . 0 0 0 0 0 0 0 1 . 0 0 0 0 0 0 0 0
172      .      16      .      1      .      0
0 0 0 0 0 0 0 0 . 0 0 0 0 0 0 0 0 . 0 0 0 0 0 0 0 0 . 0 0 0 0 1 1 1 1
0      .      0      .      0      .      15

```

Wildcard masks

- '0'
- '1'

No match! EIGRP will **not** be activated on the interface.

```
10101100 . 00010000 . 00000001 . 00001110
172      .    16      .     1       .    14
```

EIGRP network command:

```
10101100 . 00010000 . 00000001 . 00000000
172      .    16      .     1       .     0

00000000 . 00000000 . 00000000 . 00000111
0        .     0      .     0       .     7
```

Wildcard masks

- '0'
- '1'

Match! EIGRP will be activated on the interface.

```
10101100 . 00010000 . 00000001 . 00001110
172      .    16      .     1       .    14
```

EIGRP network command:

```
10101000 . 00000000 . 00000000 . 00000000
168      .     0      .     0       .     0

00000111 . 11111111 . 11111111 . 11111111
7        .   255     .   255     .   255
```

"show ip protocols" (for EIGRP)

```

R1#show ip protocols
*** IP Routing is NSF aware ***

Routing Protocol is "eigrp 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Default networks flagged in outgoing updates
  Default networks accepted from incoming updates
  EIGRP-IPv4 Protocol for AS(1)
    Metric weight K1=1, K2=0, K3=1, K4=0, K5=0
    NSF-aware route hold timer is 240
    Router-ID: 172.16.1.14
    Topology : 0 (base)
      Active Timer: 3 min
      Distance: internal 90 external 170
      Maximum path: 4
      Maximum hopcount 100
      Maximum metric variance 1

  Automatic Summarization: disabled
  Maximum path: 4
  Routing for Networks:
    10.0.0.0
    172.16.1.0/28
  Passive Interface(s):
    GigabitEthernet2/0
  Routing Information Sources:
    Gateway          Distance      Last Update
    10.0.12.2         90           00:00:23
    10.0.13.2         90           00:00:23
  Distance: internal 90 external 170

```

"Router ID"

ROUTER ID order of priority:

- Manual configuration
- Highest IP ADDRESS on a LOOPBACK INTERFACE

- Highest IP ADDRESS on a PHYSICAL INTERFACE

```
R1(config-router)#eigrp router-id ?
  A.B.C.D  EIGRP Router-ID in IP address format

R1(config-router)#eigrp router-id 1.1.1.1
```

"Distance" (AD)

EIGRP has TWO VALUES:

- Internal = 90
- External = 170

MEMORIZE THESE VALUES!

"show ip route" (for EIGRP)

```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
C       10.0.12.0/30 is directly connected, GigabitEthernet0/0
L       10.0.12.1/32 is directly connected, GigabitEthernet0/0
C       10.0.13.0/30 is directly connected, GigabitEthernet1/0
L       10.0.13.1/32 is directly connected, GigabitEthernet1/0
D       10.0.24.0/30 [90/3072] via 10.0.12.2, 00:11:09, GigabitEthernet0/0
D       10.0.34.0/30 [90/28416] via 10.0.13.2, 00:11:09, GigabitEthernet1/0
172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
C       172.16.1.0/28 is directly connected, GigabitEthernet2/0
L       172.16.1.14/32 is directly connected, GigabitEthernet2/0
D       192.168.2.0/24 [90/3072] via 10.0.12.2, 00:11:09, GigabitEthernet0/0
192.168.3.0/25 is subnetted, 1 subnets
D       192.168.3.0 [90/3072] via 10.0.13.2, 00:11:10, GigabitEthernet1/0
D       192.168.4.0/24 [90/3328] via 10.0.12.2, 00:11:09, GigabitEthernet0/0
```

NOTE the large METRIC numbers. This is a DOWNSIDE to EIGRP - even on small networks!

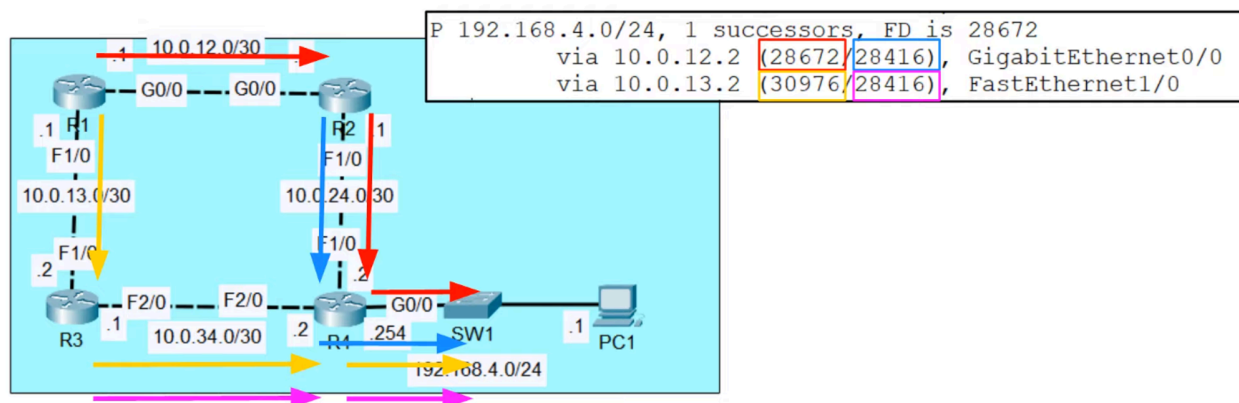
EIGRP METRIC

- By DEFAULT, EIGRP uses BANDWIDTH and DELAY to calculate METRIC
- Default "K" values are:
 - K1 = 1, K2 = 0, K3 = 1, K4 = 0, K5 = 0

💡 Simplified calculation : METRIC = BANDWIDTH (Slowest Link) + DELAY (of ALL LINKS)

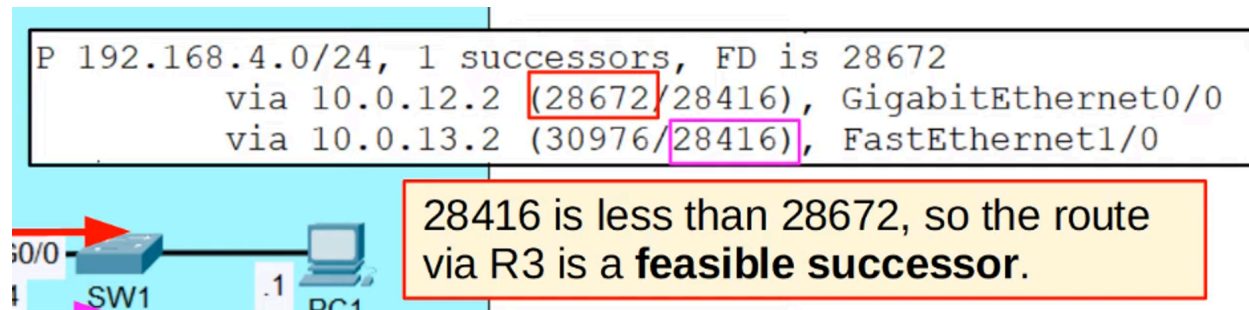
EIGRP TERMINOLOGY

- **Feasible Distance** = This ROUTER's METRIC value to the ROUTE's DESTINATION
- **Reported Distance** (aka **Advertised Distance**) = The neighbor's METRIC value to the ROUTE's DESTINATION
- **Feasible Distance** = This router's metric value to the route's destination.
- **Reported Distance** (aka Advertised Distance) = The neighbor's metric value to the route's destination.



- **Successor** = the ROUTE with the LOWEST METRIC to the DESTINATION (the best route)
- **Feasible Successor** = An alternate ROUTE to the DESTINATION (not the best route) which meets the *feasibility condition*

FEASIBILITY CONDITION : A ROUTE is considered a **Feasible Successor** if it's **Reported Distance** is LOWER than the Successor ROUTE's **Feasible distance**



EIGRP : UNEQUAL-COST LOAD-BALANCED

```
R1#show ip protocols

Routing Protocol is "eigrp 100 "
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Default networks flagged in outgoing updates
  Default networks accepted from incoming updates
  EIGRP metric weight K1=1, K2=0, K3=1, K4=0, K5=0
  EIGRP maximum hopcount 100
  EIGRP maximum metric variance 1
```

"maximum metric variance 1" = the DEFAULT value

Variance 1 = only ECMP (Equal-Cost Multiple Path) load-balancing will be performed

```
R1(config-router)#variance ?
  <1-128> Metric variance Multiplier
R1(config-router)#variance 2
```

Variance 2 = **feasible successor** routes with an FD up to 2x the **successor** route's FD can be used to load-balance

💡 EIGRP will only perform UNEQUAL-COST LOAD-BALANCING over ***feasible successor*** ROUTES. If a ROUTE doesn't meet the ***feasibility condition***, it will NEVER be selected for load-balancing, regardless of **variance**